

Ocular Manifestations

The eyelids develop increased pigmentation and loss of lashes. Atrophy of the eyelid skin can lead to ectropion, entropion, or, in severe cases, complete loss of the eyelids. Benign conjunctival inflammatory masses or papillomas of the eyelids may occur. Epithelioma, squamous cell carcinoma, and melanoma commonly affect UV-exposed portions of the eye. Ocular manifestations may be more severe in Black patients.

Neurologic System

Neurologic abnormalities occur in approximately 30% of patients. Onset may be in infancy or delayed until the second decade of life. Clinical features range from mild (e.g., isolated hyporeflexia) to severe, including progressive intellectual disability, sensorineural deafness (typically starting with high-frequency hearing loss), spasticity, and seizures.

The most severe neurologic presentation is De Sanctis-Cacchione syndrome, which includes the classic cutaneous and ocular features of xeroderma pigmentosum (XP) along with neurologic and somatic abnormalities such as: - Microcephaly - Progressive cognitive decline - Low intelligence - Hyporeflexia or areflexia - Choreoathetosis - Ataxia - Spasticity - Achilles tendon shortening leading to eventual quadriplegia - Dwarfism - Delayed or incomplete sexual development

The complete De Sanctis-Cacchione syndrome is rare, but many XP patients exhibit one or more of these neurologic features. In clinical practice, deep tendon reflex testing and routine audiometry are useful screening tools for neurologic involvement. When early neurologic abnormalities are suspected, brain MRI may reveal enlarged ventricles.

Autopsy findings in patients with neurologic symptoms show predominant loss (or absence) of neurons, particularly in the cerebrum and cerebellum. Evidence supports primary axonal degeneration as a key neuropathologic mechanism.

Laboratory Abnormalities

Cellular Hypersensitivity

Cultured cells from XP patients typically grow normally in the absence of DNA-damaging agents. However, after UV exposure, their population growth rate and single-cell colony-forming ability are significantly reduced compared to normal cells. Fibroblasts from XP patients show a wide range of post-UV survival, from extremely low to nearly normal (see Chap. 109).

XP fibroblasts are deficient in repairing UV-damaged viruses or plasmids to a functionally active state. Host cell reactivation assays have identified this repair defect in all tested forms of XP.

UV-irradiated XP fibroblasts exhibit hypermutability relative to normal fibroblasts. This increased mutagenesis is believed to underlie the high frequency of sunlight-induced somatic mutations that drive skin and ocular cancers in XP patients.

Chromosome Abnormalities

XP cells generally have a normal karyotype without spontaneous chromosomal instability (unlike Bloom syndrome). However, after UV exposure, they show abnormally increased chromosome breakage and sister chromatid exchanges. The extent of this UV-induced chromosomal damage varies among patients.

DNA Repair

In 1968, Cleaver reported that the hypersensitivity of XP cells to UV radiation results from defective DNA repair. He demonstrated impaired UV-induced repair replication, indicating a defect in the nucleotide excision repair (NER) pathway. Most XP cells respond normally to x-rays, highlighting the specificity of the defect to UV-induced DNA damage.

The genes responsible for the seven NER-defective complementation groups (XP-A to XP-G) and the XP variant have been cloned, and their functions are under investigation (see Chap. 109).

Complementation Groups

Genetic heterogeneity among XP patients was identified by fusing fibroblasts from different individuals and assessing complementation (see Chap. 109). As of 2006, seven DNA excision repair-deficient complementation groups (XP-A to XP-G) have been defined, and their corresponding genes identified (see Fig. 140-1).

Additional patients with clinical XP but intact NER are classified as XP variants. These cells show mild UV hypersensitivity, particularly evident as increased inhibition of cell growth after UV exposure, which is enhanced by caffeine.